Comparative Effects of Frying, Boiling and Curing on Lipid Oxidation in Chicken Meat Using the TBARS Assay

[*1] Inegbedion A., [2]Aguebor-Ogie N.B.

[1]Department of Medical Biochemistry, College of Medicine, Ambrose Alli University, Ekpoma, Edo State, Nigeria.

[2]Department of Medical Biochemistry, School of Basic Medical Sciences, University of Benin, Benin City, Nigeria.

ABSTRACT

Chicken meat is an important source of high-quality protein and essential nutrients; however, thermal processing and curing may promote lipid oxidation, thereby affecting its nutritional quality and shelf stability. This study evaluated the effects of boiling, frying, and curing on lipid oxidation in chicken meat using malondialdehyde (MDA) concentration as an indicator of oxidative deterioration. Fresh chicken meat samples were processed into five treatment groups: raw, raw-cured, boiled, fried, and boiled–fried cured. Lipids were extracted using the Folch method, and lipid oxidation was quantified by the thiobarbituric acid reactive substances (TBARS) assay, with absorbance measured spectrophotometrically at 532 nm. The results demonstrated that frying and curing significantly increased ($P < 0.05$) MDA concentrations compared with boiled samples, indicating greater lipid peroxidation following these processing methods. In contrast, boiling produced the lowest level of lipid oxidation, suggesting that it better preserves lipid stability during processing. These findings indicate that processing methods markedly influence the oxidative stability of chicken meat and that high-temperature frying and curing may accelerate lipid deterioration. Boiling may therefore be a preferable processing method for minimizing lipid oxidation and preserving the nutritional quality of chicken meat

Key words: Chicken meat, Lipid oxidation, Malondialdehyde, TBARS assay, Boiling, Frying, curing.

***Corresponding author.** Tel. +2348075358250: **E-mail address:** mails4augusta@yahoo.com

1. INTRODUCTION

Poultry meat, particularly chicken, is one of the most widely consumed sources of animal protein worldwide because of its high nutritional value, affordability, and versatility. It provides high-quality protein, essential amino acids, vitamins, and minerals while containing relatively low levels of fat and connective tissue compared with red meat, making it an important component of healthy diets [1, 2]. In addition to its nutritional benefits, the lipid fraction of chicken meat contributes significantly to flavour, juiciness, texture, and the absorption of fat-soluble vitamins. However, these lipids, especially polyunsaturated fatty acids, are highly susceptible to oxidative deterioration during processing and storage [3].

Cooking is an essential step in meat preparation because it improves palatability, enhances digestibility, inactivates pathogenic microorganisms, and extends shelf life [4, 5]. Nevertheless, different processing methods expose meat to varying thermal and chemical conditions, which may influence its nutritional composition and oxidative stability. Common processing methods such as boiling, frying, and curing differ considerably in their mechanisms of heat transfer and preservation, thereby affecting the extent of lipid oxidation and overall meat quality. Boiling involves cooking meat in water at approximately 100°C and is generally regarded as a relatively mild thermal

treatment. Although boiling may result in the loss of some water-soluble nutrients into the cooking medium, it is less likely to expose lipids to the extreme temperatures that accelerate oxidative degradation [6]. In contrast, frying involves cooking meat in hot oil at temperatures typically ranging from 160°C to 190°C. The prolonged exposure of edible oils and meat lipids to high temperatures promotes hydrolysis, oxidation, and polymerisation reactions, leading to the formation of lipid oxidation products that may adversely affect flavour, nutritional quality, and shelf stability [3, 7]. Curing, which commonly involves the application of salt, nitrites, sugars, and other curing agents, enhances preservation and sensory characteristics but may also influence oxidative processes depending on the curing formulation and storage conditions [8-10].

Lipid oxidation is one of the principal biochemical processes responsible for quality deterioration in meat products. It involves the oxidative degradation of unsaturated fatty acids, resulting in the formation of aldehydes, ketones, and other secondary oxidation products that contribute to rancid flavours, undesirable odours, discoloration, reduced nutritional value, and shortened shelf life [11, 12]. Furthermore, excessive accumulation of lipid oxidation products has been associated with potential adverse health effects because some oxidation-derived compounds exhibit cytotoxic and pro-inflammatory properties [13, 14].

Among the various indicators of lipid oxidation, malondialdehyde (MDA) is widely recognised as one of the most reliable secondary products of lipid peroxidation. Quantification of MDA using the thiobarbituric acid reactive substances (TBARS) assay is one of the most widely employed analytical methods for evaluating oxidative deterioration in meat and meat products because of its simplicity, sensitivity, and reproducibility [15, 16]. Measurement of MDA provides valuable information on the extent of oxidative damage resulting from different processing and storage conditions.

Although numerous studies have investigated lipid oxidation in meat products, the extent to which common processing methods such as boiling, frying, and curing influence oxidative deterioration in chicken meat remains incompletely understood, particularly under locally adopted processing conditions. Comparative evaluation of these processing techniques is important because differences in lipid oxidation may influence the nutritional quality, sensory acceptability, and storage stability of chicken meat.

Therefore, the present study aimed to evaluate the effects of boiling, frying, and curing on lipid oxidation in chicken meat by determining malondialdehyde concentrations using the TBARS assay following lipid extraction by the Folch method. The findings of this study are expected to contribute to a better understanding of the effects of common processing methods on the oxidative stability of chicken meat and provide evidence to support healthier food preparation practices and improved meat quality management.

2. MATERIALS AND METHODS

2.1. Chemicals and reagents: All chemicals and reagents used in this study were of analytical grade. Chloroform, methanol, thiobarbituric acid (TBA), butylated hydroxytoluene (BHT), glacial acetic acid, and other laboratory reagents were obtained from standard commercial suppliers and used without further purification.

2.2. Sample collection: Fresh chicken meat (*Gallus gallus domesticus*) was purchased from Uselu Market, Ovia North-East Local Government Area, Edo State, Nigeria. Fresh ingredients used for curing, including garlic (*Allium sativum*), ginger (*Zingiber officinale*), onion (*Allium cepa*), common salt (sodium chloride), and groundnut oil for frying, were purchased from the same market.

2.3. Sample preparation: The chicken meat was trimmed to remove the skin, bones, and visible connective tissue before being washed thoroughly with potable water. The meat was cut into approximately equal-sized portions and randomly assigned to five treatment groups: Group 1: Raw sample (RS), Group 2: Raw cured sample (RCS), Group 3: Boiled sample (BS), Group 4: Fried sample (FS) and Group 5: Boiled–fried cured sample (BFCS). Each treatment was prepared under identical laboratory conditions to minimise experimental variation.

Curing procedure: A dry-curing method was employed. The curing mixture consisted of finely blended garlic, ginger, onion, and common salt. The seasoning mixture was uniformly rubbed onto the surface of the designated chicken samples and allowed to stand for an appropriate curing period before further processing.

Boiling procedure: Samples assigned to the boiled group were cooked in distilled water at approximately 100°C for 30 minutes. After boiling, the samples were allowed to cool to room temperature before lipid extraction.

Frying procedure: Groundnut oil (1 L) was heated in a frying pan to 160–190°C. Chicken samples were immersed completely in the heated oil and fried until golden brown. Following frying, excess oil was drained, and the samples were cooled to room temperature before analysis.

2.4. Lipid Extraction: Lipids were extracted using the method described by Folch et al. [17]. Briefly, 5 g of each processed sample was homogenised with 100 mL of chloroform–methanol (2:1, v/v). The homogenate was mixed thoroughly, after which 20 mL of distilled water was added to facilitate phase separation. The mixture was centrifuged at 4,000 rpm for 5 minutes, and the lower organic phase containing the extracted lipids was carefully collected into clean glass tubes for subsequent analysis.

2.5. Determination of lipid oxidation (TBARS Assay): Lipid oxidation was assessed by measuring malondialdehyde (MDA) concentrations using the thiobarbituric acid reactive substances (TBARS) assay according to the method of Zeb and Ullah [18].

2.5. Statistical analysis: Data were analysed using IBM SPSS Statistics version 24.0 (IBM Corp., Armonk, NY, USA). Results were expressed as mean ± standard deviation (SD) (or mean ± standard error of the mean (SEM) if SEM was intentionally reported). Comparisons among the five treatment groups were performed using one-way analysis of variance (ANOVA) followed by Tukey's multiple comparison post hoc test where significant differences were detected. Statistical significance was accepted at $P < 0.05$.

RESULTS

The effects of different processing methods on lipid oxidation in chicken meat, as determined by thiobarbituric acid reactive substances (TBARS), are presented in Table 1. The raw chicken meat (control) exhibited the lowest TBARS value of 0.36 ± 0.02 μmol/g tissue weight. All processing methods resulted in increased TBARS levels relative to the control, indicating that boiling, frying, and curing promoted lipid oxidation to varying degrees. Among the individual processing methods, the boiled sample recorded the lowest TBARS value (0.48 ± 0.02 μmol/g tissue weight), representing the least degree of lipid peroxidation. The raw-cured sample showed a higher TBARS value (0.65 ± 0.03 μmol/g tissue weight), while the fried sample exhibited a significantly higher level (0.83 ± 0.04 μmol/g tissue weight) than the boiled sample ($P < 0.05$). The boiled–fried cured sample recorded the highest TBARS value (0.95 ± 0.05 μmol/g tissue weight), demonstrating the greatest extent of lipid oxidation among all treatment groups.

Overall, TBARS values increased in the order: Raw sample < Boiled sample < Raw-cured sample < Fried sample < Boiled–fried cured sample. These findings indicate that thermal processing and curing significantly influenced lipid oxidation in chicken meat, with frying and the combined boiling–frying–curing treatment producing substantially greater oxidative deterioration than boiling alone.

Table 1. Level of lipid oxidation for each group

Group	TBARS level (µmol/gm tissue weight)
Raw sample (RS)	0.36 ± 0.02
Raw cured sample (RCS)	0.65 ± 0.03
Boiled sample (BS)	0.48 ± 0.02
Fried sample(FS)	0.83 ± 0.04*
Boiled, fried and cured sample (BFCS sample)	0.95 ± 0.05*

Values are expressed as Mean ± SEM. *significant compared with control.

DISCUSSION

This study evaluated the effects of boiling, frying, and curing on lipid oxidation in chicken meat using malondialdehyde (MDA), measured by the thiobarbituric acid reactive substances (TBARS) assay, as an indicator of lipid peroxidation. The findings demonstrated that all processing methods increased MDA concentrations relative to the raw control, indicating that thermal processing and curing promoted oxidative deterioration of lipids in chicken meat. However, the magnitude of oxidation varied among processing methods, with boiling producing the lowest MDA concentration and the combined boiled–fried cured treatment producing the highest.

Lipid oxidation is one of the principal mechanisms responsible for quality deterioration in meat products because it adversely affects flavour, colour, texture, nutritional quality, and storage stability [11, 12]. Chicken meat is particularly susceptible to oxidative degradation because it contains relatively high concentrations of polyunsaturated fatty acids, which readily undergo free radical-mediated oxidation during processing and storage [19, 20]. The increase in MDA observed in the processed samples therefore reflects the oxidative changes induced by the various processing techniques.

The significantly lower MDA concentration observed in the boiled samples suggests that boiling exerted the least oxidative effect on the lipids present in chicken meat. Although boiling involves heat treatment, the cooking temperature is limited to approximately 100°C and the aqueous environment may reduce direct exposure of lipids to oxygen compared with dry-heat methods. Furthermore, some melted fat may leach into the cooking water, thereby reducing the amount of lipid available for oxidation [21, 22]. Similar findings have been reported by Domínguez et al. [23], who observed lower lipid oxidation in boiled meat than in meat subjected to higher-temperature cooking methods. These findings suggest that boiling may better preserve the oxidative stability of chicken meat than frying or combined processing methods.

Frying produced significantly higher MDA concentrations than boiling, indicating greater lipid peroxidation under high-temperature cooking conditions. Frying exposes both endogenous meat lipids and cooking oil to temperatures that often exceed 170°C, accelerating the formation of reactive oxygen species and lipid-derived free radicals [24]. In

addition, repeated heating and oxidation of edible oils may generate secondary oxidation products that are absorbed into the meat during frying, further increasing oxidative deterioration [25]. Comparable increases in lipid oxidation following frying have been documented by Falowo et al. [20] and Domínguez et al. [23], who reported significantly higher TBARS values in fried meat products than in boiled or steamed samples.

The cured samples also exhibited elevated MDA concentrations relative to the control. Curing alters the physicochemical properties of meat through the addition of salt and other curing ingredients, which may influence lipid oxidation depending on the curing formulation and storage conditions. Sodium chloride has been shown to accelerate lipid oxidation by disrupting cellular membranes and increasing the catalytic activity of pro-oxidant metal ions released from muscle proteins [8, 26]. Consequently, although curing improves flavour, preservation, and sensory quality, it may also increase oxidative deterioration if appropriate antioxidant systems are not incorporated into the curing process [27].

The boiled–fried cured samples recorded the highest MDA concentrations among all treatment groups, suggesting that sequential processing exerted additive or cumulative oxidative effects. The combination of curing, thermal denaturation during boiling, and subsequent exposure to high frying temperatures likely increased the susceptibility of membrane lipids to oxidation, resulting in greater production of secondary lipid oxidation products such as malondialdehyde. Similar cumulative effects have been reported in studies evaluating multiple processing treatments, where repeated exposure to heat and oxygen resulted in progressively higher TBARS values [20, 23]. The present findings therefore indicate that combinations of processing methods may have a greater impact on lipid stability than individual processing techniques.

From a nutritional perspective, the observed increase in lipid oxidation following frying and curing may reduce the sensory and nutritional quality of chicken meat. Oxidative degradation of polyunsaturated fatty acids can result in the loss of essential fatty acids and fat-soluble vitamins while producing aldehydes and other volatile compounds responsible for rancid flavours and undesirable odours [11, 12]. However, it should be emphasised that the present study measured only MDA concentrations and did not evaluate microbial quality, toxicological safety, sensory acceptability, or consumer health outcomes. Therefore, direct conclusions regarding adverse health effects or food safety cannot be drawn solely from the observed TBARS values.

The findings of this study support previous reports that processing method is an important determinant of lipid oxidative stability in meat products. Nevertheless, lipid oxidation is influenced by numerous interacting factors, including the fatty acid composition of the meat, duration and temperature of cooking, oxygen availability, storage conditions, antioxidant content, and the type of cooking oil used [12, 28]. Consequently, the observed differences among treatments are likely to reflect the combined influence of these factors rather than temperature alone.

The study has some limitations that should be acknowledged. The investigation assessed lipid oxidation using only the TBARS assay, which, although widely accepted, measures primarily secondary oxidation products and may not fully characterise the complex process of lipid oxidation. Additional analyses, including peroxide value, anisidine value, conjugated dienes, or volatile aldehyde profiling, would provide a more comprehensive assessment of oxidative deterioration. Furthermore, the study did not determine changes in fatty acid composition, moisture content, cooking yield, or sensory attributes, all of which are important indicators of meat quality. The inclusion of these

parameters in future studies would strengthen understanding of the effects of processing methods on chicken meat quality.

Overall, the findings demonstrate that common processing methods differ considerably in their effects on lipid oxidation. Among the treatments evaluated, boiling produced the lowest degree of lipid peroxidation, whereas the combination of curing and frying produced the greatest oxidative deterioration. These findings suggest that processing conditions should be carefully considered during meat preparation to minimise oxidative changes and preserve the nutritional and physicochemical quality of chicken meat.

5. CONCLUSION

In conclusion, all processing methods increased lipid oxidation compared with raw chicken meat. Boiling was associated with the lowest level of oxidative deterioration among the processed samples. Frying produced significantly greater lipid oxidation than boiling. The combined boiled–fried cured treatment resulted in the greatest oxidative deterioration, indicating cumulative effects of sequential processing. The order of lipid oxidation appears to be: Raw < Boiled < Raw-cured < Fried < Boiled–fried cured.

REFERENCES

1. Debut, M., Berri, C.M., Baeza, E., Sellier, N., Arnould, C., Guemene, D., Jehl, N., Boutten, B.,Jego, Y., Beaumont, C., Le Bihan-Duval, E. Variation of chicken technological meat quality in relation to genotype and preslaughter stress conditions. *Poultry Science*, 2008; 87 (12):2557-2565.
2. Petracci, M. Bianchi, M., Cavani, C., Gaspari, P., Lavazza, A. Preslaughter mortalityin broiler chickens, turkeys and spent hens under commercial slaughtering. *Poultry Science,* 2009; 88 (6);1259-1266.
3. Pearson, A.M. and Gillett, T.A. Processed meats(3rd ed.). New york, NY: Sprionger. 1999; Pp 448.
4. James, S.J., James, C. Cooking of meat: Heat processing Methods. In C. Devine and M. Dikeman (Eds.), Encyclopedia of Meat Sciences. 2nd Ed. 2014; Pp. 385-390).
5. Toldra, F. Lawrie's Meat Science (9th.ed.). Woodhead publishing. See chapter 7, The Storage and Preservation of Meat: 1-Thermal Technologies. 2023; Pp. 219 -244.
6. Aberle, E.D. Forrest, J.C. Gerrard, D.E., Mills, E.W. Principles of meat science (5th ed.)Kendall Hunt publishing company. 2012; Pp 117-198.
7. Xu, X. Lipids. In: Fennema, O.R. (ed.), Food chemistry (3rd ed.). Marcel Dekker. New York. 1999; Pp 157-223.
8. Toldra, F. Lawrie's meat science, nutrition, health and consumer aspects. *Journal of food Engineering,* 2017; 56 (2-3): 143-152.
9. Toldra, F. Hand book of meat processing (3rd ed.). Boca raton, FL CRC press/ Taylor and Francis Group. 2022; Pp 125-141.
10. Toldra, F. Hand book of Meat processing. Wiley-Blackwell. 2010; Pp 125-141
11. Frankel, E. N. Lipid Oxidation (2nd Ed.). Woodhead publishing. 2005; Pp 1-25.
12. Shahidi, F., Zhong, Y. Lipid oxidation and improving the oxidative stability. *Chemical Society Reviews,* 2010; 39:4067-4079.
13. Papuc, C., Goran, G.V. Predescu, C.N., Nicorescu, V. Mechanisms of oxidative processes in meat and toxicity induced by postprandial degradation products: A review. *Comprehensive Reviews in Food Science and Food Safety*, 2017; 16(1):96-123.
14. Zhou, B. Luo, J. Quan, W. Lou, A., Shen, Q. Antioxidants activity and sensory quality of bacon. *Foods*, 2022; 11:236.
15. Min, B., Ahn, D.U. Lipid oxidation in meat and meat products. In: Applied Muscle Biology and meat Science. 2009; Pp. 227-246.
16. Ayala, A., Munoz, M.F., Arguelles, S. Lipid peroxidation: Production, metabolism and signaling mechanisms of malondialdehyde and 4-Hydroxy-2-Nonenal. *Oxidative Medicine and Cellular Longevity*, 2014; Article ID 360438.
17. Folch, J., Lees, M., Sloane Stanley, G.H. Asimple method for isolation and purification

of total lipids from animal tissues. *Journal of Biological Chemistry,* 1957; 226(1): 497-509.
18. Zeb, A., Ullah, F. A simple spectrophotometric method for the determination of Thiobarbituric acid reactive substances (TBARS) in fried fast foods. *Journal of Analytical Methods in Chemistry*, 2016; pp 1-5.
19. Wood, J.D., Enser, M., Fisher, A.V., Nute, G,R., Sheard P.R, Richardson, R.L., Hughes, S.I., Whittington, F.M. Fat deposition, fatty acid composition and meat quality. A review. *Meat Science*, 2008; 78(4): 343-358.
20. Falowo, A.E., Fayemi, P.O., Muchenje, V. Natural antioxidants against lipid-protein oxidation deterioration in meat and meat products: A review. *Food Research International*, 2014; 64: 171-181.
21. Pathare, P.B., Roskilly, A.P. Quality and energy evaluation in meat cooking. *Food Engineering Reviews*, 2016; 8(4): 435 – 447.
22. El-Rammouz, R., Borno, A., Berrada, M., Ragot, P., Trystam, G. Effecyt of different cooking methods of rabbit meat on topographical changes, physiochemical characteristics, fatty acids profile, microbial quality and sensory attributes. *Meat Science*, 2021; 181: 108612.
23. Dominguez, R., Pateiro, M., Gagaoua, M., Barba, F.J., Zhang, W., Lorenzo, J.M. A comprehensive review on lipid oxidation in meat and meat products. *Antioxidants*, 2019; 8 (10):429: 1-31.
24. Choe, E., Min, D.B. Mechanisms and factors for edible oil oxidation. *Comprehensive Reviews in Food Science and Food Safety*, 2007; 6(4):169-186.
25. Saguy, I.S., Dana, D. Integrated approach to deep fat frying: Engineering nutrition, health and consumer aspects. *Jounal of Food Engineering,* 2003; 56 (2-3): 143-152.
26. Honikel, K.O. The use and control of nitrate and nitrite for the processing of meat products. *Meat Science,* 2008; 78 (1-2): 68-76.
27. Shah, M.A., Bosco, S.J.D., Mir, S,A. Effects of curing, fermentation and processing on the quality characteristics of meat products: A review. *Journal of Food Processing and Technology,* 2014; 5(7): 1-9.
28. Choe, E., Min, D.B.Chemistry of deep fat frying oils. *Journal of Food Science*, 2007; 72 (5): R77-R86.